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A machine learning framework for predicting long-term graft survival after kidney transplantation

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ABSTRACT

Kidney transplantation (KT) is an optimal treatment for end-stage renal disease (ESRD). Currently, short-term KT outcomes are indeed excellent, but long-term successful outcomes are still difficult to achieve, and improving them is crucial for kidney recipients. An early and accurate prediction of long-term graft survival helps healthcare practitioners to create a more personalized treatment plans for patients and facilitates the performance of clinical trials. In this study, we propose a machine learning framework to early predict graft survival after five years of KT and determine the most influential parameters that affect the survival. Our dataset was collected from Charles Nicolle Hospital in Tunis in Tunisia and it included pre, peri, post KT aspects. We utilized four machine learning algorithms to select the most important features: the least absolute shrinkage and selection operator logistic regression (Lasso-LR), Random Forrest (RF), Decision Tree (DT), and Chi-square (Chi-sq). We utilized three Scikit-learn functions to implement those algorithms: SelectFromModel (SFM), Recursive Feature Elimination (RFE), and SelectKBest (SKB). Five algorithms were utilized to builds prediction models based on the data groups resulted from the feature selection step: logistic regression (LR), k-nearest neighbors (KNN), extreme gradient boosting (XGB), and artificial neural network (ANN). We evaluated the models using five performance measures: accuracy, sensitivity, specificity, F1 measure, and area under the curve (AUC). XGBoost resulted the best model with the highest AUC (89.7%). It was based ten features selected by RF algorithm and SFM function. The accuracy, sensitivity, specificity, and F1 of the best model were 91.5%, 91.9%, 87.5%, and 89.6%, respectively. This study proposes a novel approach for investigating long-term allograft survival while considering the complex relationship between all KT aspects and long-term outcomes. Our framework can be used as a decision support system for Nephrologists to early detect graft status, which helps in developing safer recommendations for kidney patients and consequently obtaining positive KT outcomes and mitigating the risks of graft failure.

1. Introduction

Chronic kidney disease is a global health burden as it currently affects about 850 million people worldwide (Jager, Kovesdy, Langham, Rosenberg, & Jha, 2019). It is a major risk factor that leads to irreversible kidney failure, or what is so called the end-stage renal disease (ESRD). After kidney failure, a patient needs renal replacement therapy either by dialysis or kidney transplantation (KT). The latter treatment is more desirable and cost-effective as it improves the quality of life and survival rates of patients with severe chronic kidney disease (Abecassis et al., 2008). According to the Global Observatory on Donation and Transplantation (GODT), about 100,000 KT are performed per year around the world (transplant-observatory.org). Despite the high number of KT, the number of people on waiting lists continues to grow. In 2019, there were more than 113 k people on the U.S. national kidney transplant waiting list with an average of one person was added every 10 min. Approximately, 20 people die every day while waiting for a kidney transplant (americantransplantfoundation.org). Kidney allograft failure

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is a fatal outcome that contributes to the backflow of people to an already-overburdened waitlist and improving KT survival is therefore necessary.

KT outcomes can be divided into short-term and long-term. Short-term is referred to the results of KT within the first year (e.g., acute rejection), while long-term outcomes are referred to the follow-up results of KT after one year or more (e.g., graft failure) (Legendre, Canaud, & Martinez, 2014). Short-term outcomes have been significantly improved due to the advancement of surgical techniques and immunosuppression drugs. However, there has been no significant improvement achieved regarding long-term outcomes (Meier-Kriesche, Schold, & Srinivas, 2004; Coemans et al., 2018; Rana & Godfrey, 2019). This paper focuses on predicting graft survival after five years of KT using pre, peri, and early post-transplant features. The reason of selection five years as the end of follow-up in our study is the trade-off between the size of the data and the length of the follow-up period. If we choose long follow-up period (e.g., ten years), the number of patients records with available follow up will be reduced significantly, which will affect the accuracy of our study. On the other hand, if we choose short follow-up period, this will affect the minimize the benefit of the proposed model. Predicting long-term KT outcomes enables nephrologists to understand the progression of graft function of each patient and consequently design a more personalized treatment plan and clinical care. It also helps in evaluating the performance of clinical trials that are aimed to assess the long-term effects of immunosuppression drugs on graft survival. Over the past 20 years, biopsy-proven rejection and one-year graft survival were the main criteria used by drug regulatory agencies to approve immunosuppression drugs for KT (Stegall, Morris, Alloway, & Mannon, 2016). These endpoints are not enough to improve long-term graft survival. In other words, developing safer immunosuppression drugs requires clinical trials to be conducted over a long time period, which is not cost-effective for pharmaceutical industries. This leads to the need for an alternative tool that can predict long-term graft survival (Stegall et al., 2016).

Utilizing machine learning is an excellent option to predict long-term KT outcomes. However, we still lack a comprehensive predictive model that can assess long-term KT outcomes based on a patient's information, KT procedure attributes, and other short-term KT aspects. In addition, the diversity and complexity of the factors influencing KT outcomes make the assessment of graft failure at an individual patient level more difficult (Legendre et al., 2014). Machine learning algorithms can build accurate prediction models comparing with other traditional statistical methods such Kaplan–Meier estimator and Cox regression. Machine learning approaches can capture complex and nonlinear relationships among variables (Chen & Asch, 2017). They help in avoiding preselected improper approximation functions, overcoming the problems of discrete and continuous factors, dealing with incomplete and noisy data. They are also efficient in handling data with large number of features. However, most machine learning algorithms have many parameters, and finding the optimal values of those parameters needs many experiments.

In this paper, we propose a comprehensive machine learning framework for predicting graft survival after five years of KT. We collect KT data for more than 800 patients between 1986 and 2019 from Charles Nicolle's Hospital, which is in Tunis in Tunisia. Then, we utilize random forest (RF), decision tree (DT), Chi-square (Chi-sq), and the least absolute shrinkage and selection operator logistic regression (lasso-LR) to select the best features that explain long-term graft survival. The four algorithms are implemented in Python using three Scikit-learn functions: SelectFromModel (SFM), recursive feature elimination (RFE), and SelectKBest (SKB). The best features resulted from each algorithm are used to build five predictive models using logistic regression (LR), k-nearest neighbors (KNN), extreme gradient boosting (XGB), DT, and artificial neural network (ANN). The model with the best performance is considered as the final and recommended model. The contribution of this study is as follows: 1) We build a comprehensive predictive framework for understanding long-term graft survival after KT; 2) We consider most of the features that affect the kidney transplant failure by taking into account pre, peri, and post KT information to build our models, which makes our study more practical and robust; 3) Up to our knowledge, the best model we obtain outperforms all the machine learning models that were proposed in the literature for predicting long-term outcomes of KT, which will be explained in Section 4 in details.

The remainder of this paper is as follows: Section 2 reviews the relevant studies, while Section 3 presents the data collection and the methodologies utilized in this study. The results of our study are presented and discussed in Section 4. The conclusion and future are presented in in Section 5.

2. Related work

Machine learning has been applied for predicting different KT outcomes. Some studies predicted KT outcomes at single time point. Brier, Ray, and Klein (2003) and Decruyenaere et al. (2015) predicted delayed graft function within the first week of KT using ANN, RF, DT, and linear and radial support vector machines (SVM). Shaikhina et al. (2019) predicted acute anti-body mediated rejection 30 days post KT using RF and DT. Other studies predicted graft failure at different points of time after KT, which was defined by either returning a patient to dialysis or preemptive KT (Brown et al., 2012). Other studies developed models for predicting the combined outcome of graft failure and patient death (Lin, Horn, Hurdle, & Goldfarb-Rumyantzev, 2008). In our study, we consider only graft failure (death censored), since graft survival and patient survival may have different predictors. Topuz, Zengul, Dag, Almehmi, and Yildirim (2018) categorized kidney graft survival into three risk levels: high risk (less than one year), medium risk (1–5 years), and low risk (more than five years). They used Bayesian belief network for modeling. The main limitation of their study is that they included only pre-transplant features. They also used data of KT from only deceased donors, which makes their model non-usable predicting graft survival of KT from living donors, knowing that most of KT cases come from living donors (Wu et al., 2017). In our study, we aim to predict graft survival at a single time point (five years), based on pre, peri, and post-transplant features. Furthermore, our data was collected from both living and deceased donors, which strengthens the generalizability of our study.

Various features were considered in the literature for predicting KT outcomes. The number of predictors considered ranged from 6 (Lasserre, Arnold, Vingron, Reinke, & Hinrichs, 2012) to 71 (Lin et al., 2008). These features can be divided into three groups: pre-transplant, *peri*-transplant, and post-transplant features. Some studies considered only pre-transplant variables (Goldfarb-Rumyantzev, Scandling, Pappas, & Smout, 2003; Lin et al., 2008; Li et al., 2010; Brown et al., 2012; Lasserre et al., 2012; Shahmoradi, Langarizadeh, Pourmand, Fard, & Borhani, 2016; Topuz et al., 2018; Shaikhina et al., 2019), while others included variables from the three groups (Brier et al., 2003; Krikov et al., 2007; Akl, Ismail, & Ghoneim, 2008; Greco et al., 2010; Decruyenaere et al., 2015; Nematollahi, Akbari, Nikeghbalian, & Salehnasab, 2017; Yoo et al., 2017). Greco et al. (2010) used only post-transplant features to predict chronic allograft nephropathy in the fifth year. Our database includes pre-transplant, *peri*-transplant, and post-transplant features, which makes our study more comprehensive as it captures all the aspects of KT. To select important features, some studies relied on experts' opinions for selecting important features (Lin et al., 2008; Brown et al., 2012; Nematollahi et al., 2017), while others used univariate statistical tests (Akl et al., 2008). Few studies used advanced machine learning algorithms. Topuz et al. (2018) used elastic nets ANNs, bootstrap forest (BF), and SVMs to select essential predictors. In our study, before modeling, we implement various criteria eliminate unnecessary features, and then we use advanced machine learning methods to select the most important ones.

For the database size of the previous studies, there was a big variation. Some studies had large datasets (e.g., more than 30 K kidney recipients), while others had only 80 patients. More than 50% of the

previous studies had datasets with less than 1000 kidney recipients. Interestingly, the models developed with a small dataset (e.g., 80 KT) had similar accuracy compared with the models that were trained with larger datasets (Senanayake et al., 2019). Datasets were collected from deceased donors (Brier et al., 2003; Goldfarb-Rumyantzev et al., 2003; Brown et al., 2012; Decruyenaere et al., 2015; Topuz et al., 2018), living donors (Akl et al., 2008) or both (Krikov et al., 2007; Lin et al., 2008; Greco et al., 2010; Li et al., 2010; Nematollahi et al., 2017; Yoo et al., 2017; Shaikhina et al., 2019). In this study, our dataset has 800KT patients and it is collected from both living and deceased donors.

3. Material and methods

This section presents the research framework and methodologies of this paper. We explain all the steps we have conducted to build our models including data collection and preprocessing feature selection, model training and testing, and model evaluation.

4. Research framework

Fig. 1 presents the research formwork of this study. In phase 0, we review and identify the gap in the literature. The goal of this study is also defined in Phase 0. Phase I explains the data collection process, criteria for dropping unnecessary variables, and data preprocessing methods. In phase II, we visualize the features resulted from Phase I to understand them. Then, we utilize four feature selection approaches: Lasso-LR, RF, DT, and Chi-sq to select the best subset of features. Lasso-LR, RF, and DT algorithms are implemented using two different Scikit-learn functions: SFM and RFE, while Chi-sq is implemented only with SKB function. Since our data is imbalanced, all the feature selection methods are implemented on an oversampled data. Phase II results in seven different data groups from the main dataset, which come from Lasso-LR_SFM, RF_SFM, DT_SFM, Chi-sq_SKB, Lasso-LR_RFE, RF_RFE, and DT_RFE.

In phase III, each data group resulted from phase II is used to train and test five predictive models: LR, KNN, XGB, DT, and ANN. In other words, 35 fives models are built (7 data groups resulted from phase II × 5 training algorithms = 35). While building the 35 models, each data group is divided into training and testing sets, then oversampling is applied on only the training set to overcome the data imbalance problem. During the training stage, GridSearchCV Scikit-Learn function is utilized to select the best parameters of each model. GridSearchCV fits each model using all the combinations of a parameter values and evaluate each combination using K-fold cross validation. The evaluation is performed based on a training accuracy. Then, Then, the models with the best parameters are used for prediction and evaluations in phase IV. Five performance measures were utilized for the evaluation: accuracy, sensitivity, specificity, F1 measure, and AUC. The confusion matrix is also used to visualize the prediction results. The model with the best performance is considered as the final model.

4.1. Data collection and preprocessing

In this study, our data is collected from KT database of Charles Nicolle hospital, which is located in Tunis, Tunisia. This section explains our data preparation process including all the steps that have been taken to make the dataset ready for modeling.

4.1.1. Data preprocessing

The database has 800 KT performed between 1986 and 2019. Initially, more than 202 features are obtained from the hospital database, and they include information about all the aspects of KT such as recipient and donor information, pre, pri, early-post transplantation, biological monitoring after KT, histological data, and long-term evolution. To make the data ready for modeling, several inclusion and exclusion criteria are implemented. Also, the data is preprocessed before modeling. Suppose that m is the number of features and n is the number of KT patients, the following steps are taken to handle the initial dataset (See Fig. 2):

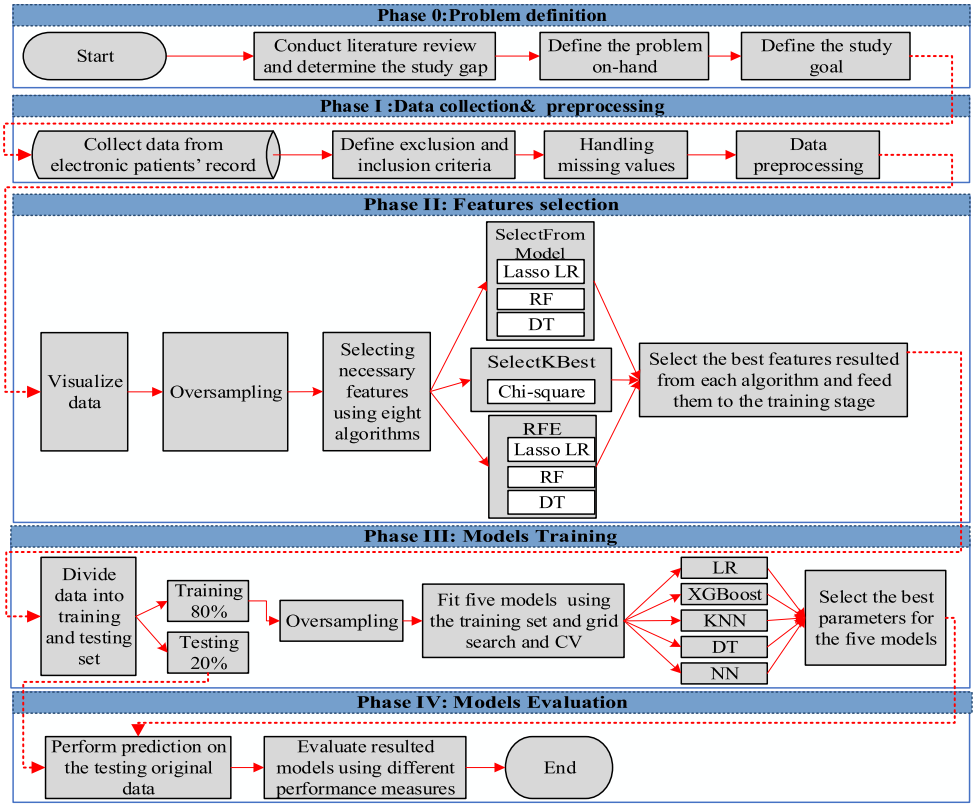


Fig. 1. Research framework.

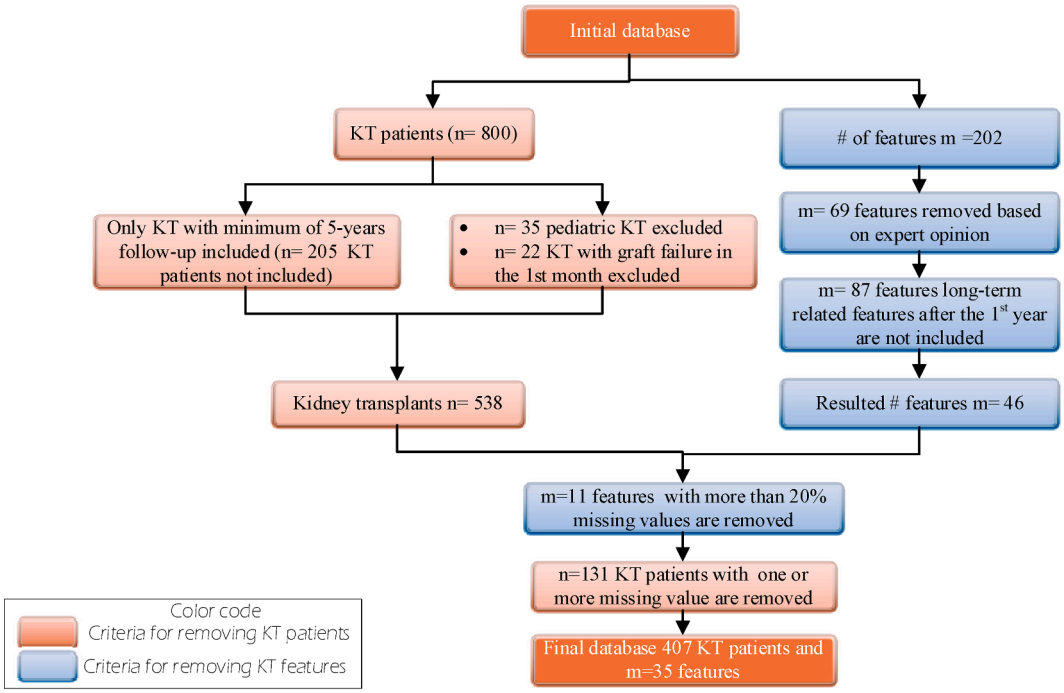


Fig. 2. Process of preparing the study dataset.

- We remove $m = 69$ irrelevant variables based on a nephrologist's opinion. Those features are known to not have an impact on graft survival (e.g. variables concerning the surgical procedure). This reduces the number of features from $m = 202$ to $m = 133$.
- To make sure that we have five years follow-up period for all the KT cases that are used to build our models, all the transplants that were performed after December 2014 are not removed ($n = 205$). This reduces the number of KT patients from $n = 800$ to $n = 595$.
- For features that explains post-transplant events, only information about the first year after KT are included. Thus, all featuers about long-term complications such as neoplasic complications and infections occured after the first year are removed. We also remove featuers about biological monitoring such as serum creatinin level after the first year, which reduces the number of variables as follows: $m = 133 - 87 = 46$.
- We exclude $n = 35$ pediatric KT recipient, who are under 15 years old and $n = 22$ KT with graft failure during the first 30 days of KT. After this step and before handling missing data, the final number of KT patients and features are $n = 538$ and $m = 46$, respectively.
- For missing values, we firstly removed the variables that have more than 20% missing data ($m = 11$) such as hemoglobin level before KT, recipient weight, and body mass index (BMI). This step reduces the number of features from $m = 46$to $m = 35$. Finally, we remove the whole rows that contain one or more missing values, which reduced the number of patients by $n = 131$. This concludes that our final database includes 407 patients and 35 variables.

After deciding the final number of KT patients and features, the categorial features are encoded using integer encoding, where each category is given in integer number. Finally, input features are scaled from some algorithms. For example, no scaling is applied when RF, DT, XGB are the training algorithms, while for KNN, NN, and Lasso-LR, the input features are scaled between the maximum and minimum. The scaling is conducted using $MinMaxScaler().fit_transform(X)$, Scikit-learn function, where X is the input features.

4.1.2. Features considered after data preprocessing:

The final features obtained after preparing the dataset can be divided into three groups, pre, peri, and post KT. The visualization of all the features are shown in Appendix A. This section explains all of them in detail.

Pre-transplantation variables: This group includes recipient, donor, and immunological information. The recipient-related features are as follows:

Recipient information: It includes historical information about recipients as follows (Fig. A1 in Appendix A):

- **Recipient age**: This is a metric feature and it represents the age of recipients at the time of KT.
- **Recipient gender**: This is a binary feature and it denotes the gender of a recipient whether it is male or female.
- **Diabetes**: This is a binary feature to indicate whether a KT patient has diabetes or not.
- **Hypertension**: This is a binary feature to denote if a KT recipient has a history of hypertension before KT.
- **Kidney disease**: this is an integer feature and it represents the initial causes of ESRD, which are divided into six groups: glomerular nephropathy, tubulointerstitial nephropathy, vascular nephropathy, diabetic kidney disease, hereditary nephropathy, and indeterminate nephropathy.
- **Dialysis modality before KT**: This is an integer feature and it indicates renal replacement therapy modality immediately before transplant, which are hemodialysis, peritoneal dialysis, both, or preemptive KT if a patient did not have a course of dialysis before KT.
- **ERSD duration before KT (ESRD pre KT)**: This is a metric feature and it denotes the number of months of pre-transplant ESRD period.

Donor information: This category includes historical information about a donor as follows (Fig. A2 in Appendix A):

- **Donor age**: This is a metric feature and it denotes a donor age at the time of donation.
- **Donor gender**: This is a binary feature to show the gender of a donor.

- **Donor type**: This is an integer feature and it explains the status of a donor. It has three values: living donor, living, and relative to a recipient, or deceased donor.
- **Gender match**: This is an integer feature and it has four values to explain the donor-recipient gender matching: male/male, male/female, female/male, or female/female.

Immunological information: This category describes the immune-related variables for a recipient before KT as follows: **(**Fig. A3 in Appendix A)

- **Transfusion:** This is a binary feature and it shows whether a recipient has a history of red blood cell transfusion before KT.
- **Cytotoxic antibodies**: This is a binary feature and it indicates the presence or absence of HLA class I (A and B) and/or HLA class II (DR) cytotoxic antibodies in the recipient before KT.
- **Recipient blood type**: This is a four-level integer feature and it shows a recipient's blood group: O, A, B, or AB.
- **Donor blood type**: This is a four-level integer feature and indicates a donor's ABO system blood group.
- **Blood type compatibility**: This is a binary feature and it indicates the compatibility between a donor and a recipient blood group, which is either ABO identical or ABO compatible.
- **HLA A, HLA B, HLA DR mismatch**: These are three features and they are all integer. They explain the number of HLA mismatches in each HLA group (A, B, DR), which is defined as the number of HLA disparities from a recipient perspective. In each HLA gene, the number of donor alleles not shared by the recipient is calculated and it is 0, 1, or 2.

Peri-transplantation variables: This group includes data about cold and warm ischemia time and induction therapy, which are as follows (Fig. A4 in Appendix A):

- **Warm ischemia time (WIT)**: This is a metric feature and it is defined as the period between the removal of a procured kidney from an ice storage and the initiation of graft reperfusion. It is measured in minutes.
- **Cold ischemia time (CIT)**: This is a metric feature and it defines the time duration between the beginning of cold storage of a graft and reperfusion. It is measured in hours. For KT from a living donor, CIT is zero.
- **Induction**: This is an integer feature with three possible values. It represents the type of immunosuppressive initial induction therapy a patient has had. The values of the feature are either no induction or type of antilymphocyte biological agent, which takes two values: interleukin −2 receptor antagonist (anti-IL2 R) and polyclonal antilymphocyte (PAL).

Post-transplantation variables: This group involves information about immunosuppressive treatment, recipient's health status during the first year of KT, and Early post-transplant complication.

Immunosuppressive treatment: KT patients receive different treatment regimens and this category explain the ones that are included in our database as follows (See Fig. A5 in Appendix A):

- **Mmycophenolate mofetil (MMF)**: This is a binary feature and it indicates whether MMF therapy, which is an antiproliferative agent, is used after KT.
- **Azathioprine**: This is a binary feature and it indicates whether azathioprine is used as antiproliferative agent instead of MMF.
- **Tacrolimus**: This is an integer feature that explains whether tacrolimus therapy, which is a calcineurin inhibitor, is used as primary, secondary therapy or not used at all.

The recipient's health status within the first year of KT: This category explains that the health condition of kidney receipts within the first year of KT. It includes the following variables (See Fig. A6 in Appendix A):

- **Length of first hospital stay (1st hospitalization)**: This is a continuous feature that represents the number of days a patient spends at a hospital from the day of transplant procedure until discharging.
- **3-months estimated Glomerular Filtration Rate (eGFR)**: This is a continuous feature that represents kidney transplant glomerular filtration rate at the third month after KT. It is estimated by the serum creatinine clearance, which is calculated using the Modification of Diet in Renal Disease (MDRD) equation (Chen et al., 2014).
- **Number of hospital readmissions (Readmissions)**: This is an integer feature and it represents the number of hospital readmissions during the first year of KT.
- **Hypertension post KT**: This is a binary feature and it indicates whether a patient still has hypertension after KT or not.

Early post-transplant complication: This category includes the features that explains the possible complications that may happen within the first year of KT as follows (See Fig. A7 in Appendix A)

- **Delayed graft function (DGF)**: This is a binary feature and it explains whether a patient needs dialysis within the first week after KT.
- **Urological complications**: This is a binary feature that shows if a recipient experiences postoperative urological complication such as lymphocele, urine leakage, urinoma, ureteral stenosis, transplanted renal artery stenosis, and vesicoureteral reflux.
- **Infection in the first year**: This is a binary feature that denotes whether the patient experiences infectious complications (e.g., bacterial, viral, or parasitic infections) during the first-year post KT.
- **Cytomegalovirus infection (CMV)**: This is a binary feature and it explains whether a patient experiences CMV infection, which is detected by CMV serology or CMV polymerase chain reaction.
- **Acute kidney injury (AKI)**: This is a binary feature and it indicates whether a patient experiences AKI, which is an abruption decrease in kidney function.
- **Acute rejection (AR)**: This is a binary feature and it shows whether a patient experiences an acute renal allograft rejection. It is defined as an acute cellular-mediated or antibody-mediated deterioration in allograft function confirmed by biopsy or based on a strong clinical presumption according to expert opinion.

4.2. *Feature selection methods*

To select the best subset of features among the remaining 35 features, we employ LASSO-LR, RF, DT, and Chi-sq. These methods are implanted using three Scikit-learn functions. Chi-sq is implemented using the SKB function, while RF, DT, and LASSO-LR are implemented through RFE and SFM functions.

SKB function selects features based on different statistical test scores. It calculates a score for each feature and then removes all features except the K highest scoring features. In our study, SKB function is used with Chi-sq statistical test to select the best K features that have the largest effect on an output variable. For RFE function, it is considered as a wrapper feature selection method, in which a machine learning algorithm is used to calculate the score of a subset features based, then the least important features are recursively removed until a desired number of features is obtained. The scores of a subset features are calculated in different ways depending on the algorithm that is used with RFE function. When RF or DT are used with RFE, a feature importance is calculated based on how much the node that uses a feature enhances the purity of a tree across all the tree nodes. In LASSO-LR, feature importance is calculated based on the absolute values of a model's coefficients. In our study, RFE is used with three algorithms: LASSO-LR, DT, and RF.

SFM function is considered as a backward elimination method. It starts with all features and removes the least significant ones at each iteration. The model keeps removing features until no improvement is observed on the model's performance. Similar to RFE, we use SFM with LASSO-LR, RF, and DT.

4.2.1. Lasso-LR

Lasso stands for the Least Absolute Shrinkage and Selection Operator. It is considered as one of the regularization approaches that create abstract models from data with many features to avoid overfitting and reduce computation complexity. Lasso-LR has been used by many researchers as a feature selection (Wu, Yuan, & Zhuang, 2010; Algamal & Lee, 2015), classification (Kim, Kim, Jeong, Jeong, & Kim, 2018) approach. In our study, Lasso-LR is used as a feature selection method. In Lasso-LR, a penalty parameter called shrinkage penalty (λ) is multiplied by a model coefficient except for the intercept. When the value of λ is large, it shrinks some of the coefficient estimates to be exactly zero. Lasso-LR can be obtained by Eq. (1) (Hastie, Tibshirani, & Friedman, 2009).

$$l_{\lambda}^{L}(\beta) = \sum_{i=}^{N}\left[(y_i x_i \beta^T - \log(1 + e^{\beta^T x_i})\right] - \lambda \sum_{j=1}^{p} |\beta_j| \tag{1}$$

Where, x is the input variables, y is the response, β is the model coefficient matrix. We implement Lasso-LR with SFM and RFE functions.

4.2.2. Decision trees (DT)

Decision trees (DT) approach is a non-parametric algorithm that can be used for classification (Kohavi & Quinlan, 2002), regression (Xu, Watanachaturaporn, Varshney, & Arora, 2005), and feature selection (Sugumaran, Muralidharan, & Ramachandran, 2007). In our study, we use DT for feature selection and classification. It has a tree structure that starts with a single node to represent a training sample. If the sample belongs to the same class, the node becomes a leaf, otherwise, another feature is selected to be the current node of the tree. The partitioning continues until stopping criteria are met. DT utilizes different algorithms such as Iterative Dichotomiser 3 (ID3), C4.5, Classification, and Regression Trees (CART) (Zou et al., 2018). In our paper, we implement it through Scikit-learn library, which utilizes CART (*1.10. Decision Trees — scikit-learn 0.23.1 documentation*). In CART, trees are created based on the Gini index attribute, which measures how well a given feature classifies a sample into target classes. Suppose that there are n classes, Gini index for a subset D can be calculated using Eq. (2), where p_j is the percentage of the items labeled with class j in the sample set D (Han & Kamber, 2001). When DT is used for feature selection, feature importance is calculated for each feature, then depending on the function used (e.g., RFE or SFM), a feature is decided to be removed or included.

$$gini(D) = 1 - \sum_{j=1}^{n} p_j^2 \tag{2}$$

4.2.3. Random forests (RF)

Random forests (RF) approach is an ensemble method, in which learning is performed based on multiple models of several DTs to obtain better performance. In our paper, we use it as a feature selection approach. Selecting important features with RF is performed based on variable importance criterion (e.g., z-score), which evaluates how much a variable affects a classification accuracy. Suppose that there are B bootstrap samples($b = 1, 2, \cdots, B$), and T trees, which are built to calculate the importance of a variable x_j $(j = 1, 2, \cdots N)$ The steps of calculating a feature importance are as follows: 1) The classification accuracy of the out of bag (OOB) data R_b^{oob} is calculated using a classification model T_b; 2) The values of x_j is permuted with the OOB data and the classification accuracy$R_{bj}^{oob}$is calculated; 3) The mean decreases in the classification accuracy D_j for the variable x_j is calculated using Eq. (3); 4) A z-score (z_j) of the variable x_j is calculated using Eq. (4), which represents the importance of feature j, where s_j is the standard deviation of the classification accuracy (Verikas *et al.*, no date).

$$\overline{D}_j = \frac{1}{B}\sum_{b}^{B}\left(R_b^{oob} - R_{bj}^{oob}\right) \tag{3}$$

$$z_j = \frac{\overline{D}_j}{s_j/\sqrt{B}} \tag{4}$$

4.2.4. Chi-square

In this paper, Chi-sq is used as a feature selection approach. It selects features based on Chi-sq scores. The features are decided to be removed or included based on the function used for implementation, which is SKB in this paper. Suppose that there are c number of classes $(c = 1,2,\cdots,c)$, n number of samples and r number of discrete intervals $r = (i = ,1,2,\cdots r)$, then n_{ij} denotes an actual frequency of samples in the *ith* interval and *jth* class. The expected frequency of n_{ij} is denoted by μ_{ij} and can be calculated by Eq. (5). Then, the Chi-sq score can be calculated using Eq. (6). For continuous features, discretization is applied (Ma et al., 2017).

$$\mu_{ij} = \frac{\sum_{j=1}^{c} n_{ij} * \sum_{i=1}^{r} n_{ij}}{n} \tag{5}$$

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(n_{ij} - \mu_{ij}\right)^2}{\mu_{ij}} \tag{6}$$

4.3. Model development

In this paper, we utilize five algorithms to build our prediction models, which are LR, KNN, XGB, ANN, and DT. This section presents four algorithms, as DT is presented in Section 3.3.

4.3.1. Logistic regression

Logistic regression (LR) is a statistical approach that explains the relationship between multiple independent variables and one dependent variable. The independent variables can be continuous and categorical, while the dependent variable is only binary. The probabilities of the output classes are predicted based on the relationship between the input and output variables. Suppose thatx_1,x_2, $\cdots$, x_n are a model's features, n is the number of features, β_0, β_1, $\cdots$, $\beta_n$are the model coefficients. A conditional probability of each class is calculated using Eq. (7) and then the class with the highest probability is assigned (Hastie et al., 2009). In this study, we use it for classification based on the features obtained from phase II (See Fig. 1). Therefore, seven different LR models are built (See Fig. 1).

$$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_n x_n)}} \tag{7}$$

4.3.2. K-Nearest Neighbor

K-nearest neighbor (KNN) is a non-parametric algorithm and it does not make any assumptions on data distribution. Training KNN is conducted by computing the distance between an input sample and each training instance. Then a model picks k number of the closest entries in the data, then a new data point is assigned to the class that is closest to its $k$nearest neighbors. The optimal choice of the number of samples k is data dependent. Generally, As the value of k increases, the effects of noise decreases, and the classification boundaries become less distinct (Gou et al., 2019). The value k can be a user-defined (k-nearest neighbor learning) or based on the local density of points (radius-based neighbor learning) (Wang, Neskovic, & Cooper, 2006; Jalali, 2013). Various distance measures are utilized such as such as Euclidean distance, Manhattan distance and Minkowski distance or Hamming distance. Standard Euclidean distance is the most common. In this paper, Hamming distance is used since we have categorical variables. We use KNN to build

seven data groups, which are obtained from phase II (see Fig. 1).

4.3.3. Extreme gradient boosting

Extreme gradient boosting (XGB) is one of the gradient boosting machines (GBMs) methods that can be used from classification (Yu et al., 2020), regression (Chen, He, Benesty, Khotilovich, & Tang, 2015), and feature selection (Chen et al., 2020). In our paper, we used it for classification. The idea of XGB is that it builds a strong learning model from small weak learners. Generally, a single tree can be weak learning, but when multiple trees are ensembled, they can create a strong learner. In XGB, trees are generated iteratively, in which the information from a previous tree is used to enhance the quality of a current tree. A higher weight is given to the misclassified instances from previous trees. Similar to LR and KNN, XGB is used to build seven models for the data groups obtained in phase II (See Fig. 1).

4.3.4. Artificial neural network

The artificial neural network (ANN) algorithm is a machine learning approach that mimics human brain functionality. It is a collection of artificial neurons organized in layers. There are different types of ANN algorithms. In this paper, we utilize multilayer perceptron (MLP), which has three types of layers: input, output, and hidden layers (Niel & Bastard, 2019). Each layer has several nodes, in which the number of nodes in the input layer is equal to the number of input features and the number of nodes at the output layer is the same as the number of output variables (e.g., one in our paper). The hidden layer can have a variable number of nodes. The data in each layer is transformed into the next one using an activation function that introduces a degree of nonlinearity. In this paper, we use the sigmoid activation function. Similar to the previous algorithms, we use ANN to build seven models based on the data groups obtained from phase II (See Fig. 1).

4.4. Oversampling algorithm

Our dataset is imbalanced since the number of patients with a failed graft after five years (class = 0) is much less than the patients with a survived graft (class = 1) (see Fig. 3). Training a model with imbalanced data leads to poor or misleading prediction performance as the model mostly learns to predict the majority class. To overcome this problem, oversample our data using Synthetic Minority Oversampling Technique (SMOTE) approach. It is considered one of the most common oversampling algorithms (Gosain & Sardana, 2017). During the feature selection stage, we apply SMOTE on the whole dataset, while in the model development phase, we apply it only on the training set. We do not apply it on testing sets. The idea of SMOTE is that it creates new synthetic samples based on existing observations. Firstly, an observation is selected from the minority class. Secondly, *k*-nearest neighbors are found based on measures such as Euclidean distance. Finally, SMOTE creates a new instance of the minority class close to the nearest neighbors. The number of new instances created depends on the degree of SMOTE being applied. More information about SMOTE can be found in study conducted by Chawla, Bowyer, Hall, and Kegelmeyer (2002).

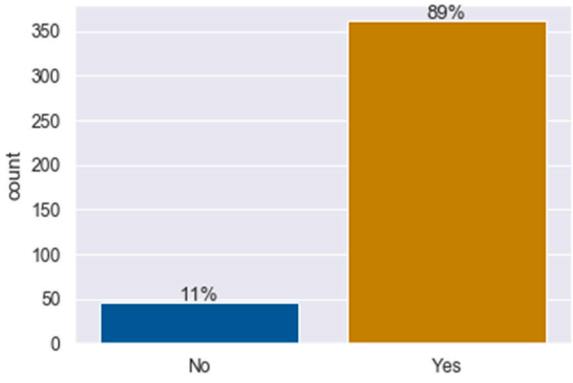


Fig. 3. Output variable – class frequency.

4.5. Performance measures

In this paper, five performance measures are used to evaluate the proposed models, which are accuracy (Eq. (8)), sensitivity (recall) (Eq. (9)), specificity (Eq. (10)), F1 score (Eq. (12)) (Han & Kamber, 2001), and area under the curve (AUC) . The confusion matrix is also used to visualize prediction performance. In our database, the class of patients whose graft survived after 5 years is labeled as class 1 (positive), and a patient with a failed graft is labeled as class 0 (negative). Therefore, a true positive (TP) is considered as an instance where a model truly predicts class 1 (graft survived), while a false negative (FN) is a case where the model incorrectly predicts class 1 (graft survived). True negative (TN) is a case where the model truly predicts class 0 (graft failed), while false positive (FP) occurs when the model misclassifies class 0 (graft failed).

Since our data is imbalanced, accuracy is misleading. Also, sensitivity and specificity are inversely proportional to each other and there is a tradeoff between them. Therefore, we consider AUC is the main measure that we utilize to decide which model is the best among the 35 models we develop. AUC is the area under all combinations of sensitivity and (1 – specificity). It is an accurate performance measure, regardless of the threshold between sensitivity and specificity.

$$Accuracy = \frac{\mathrm{TP} + \mathrm{TN}}{TP + FN + FP + FN} \tag{8}$$

$$Sensitivity = \frac{\mathrm{TP}}{TP + FN} \tag{9}$$

$$specificity = \frac{\mathrm{TN}}{TN + FP} \tag{10}$$

$$Precision = \frac{TP}{TP + FP} \tag{11}$$

$$F1 = 2^{*}\frac{Precision^{*}sensitivity}{Precision + sensitivity} \tag{12}$$

5. Results and discussion

In this section, we discuss the feature selection results and the performance of our proposed models. We also discuss the features of the best model.

5.1. Feature selection results

As mentioned previously, we utilize seven feature selection algorithms to select the best features among the 35 features. Table 1 shows the selection frequency for each feature by the seven feature selection methods in which the checkmark (✓) denotes that a feature is selected by the corresponding algorithm. The last column shows the total frequency of selection for each feature. For example, feature # 1 (MMF therapy) is selected by all the seven feature selection methods. AKI feature is also selected by the seven feature selection methods. Four features are selected four times, which are Tacrolimus, 1st hospitalization, 3-month-eGFR, and DGF. Features # 7–14 are selected by three selection methods, while features # 15–17 are selected by only two algorithms. Features # 8–26 are selected once and features # 27–35 are not selected by any algorithm. Table 1 gives a complete understanding of the features affecting graft survival. For example, features that are not selected by an algorithm do not affect graft survival comparing with the selected features. On the other hand, the features that are selected by most of the utilized algorithms are known in the literature by their

Table 1
Selection frequency of each feature.

Feature No.	Feature	Lasso SFM	DT SFM	**RF SFM**	Chi-sq SKB	DT RFE	RF RFE	Lasso RFE	Total
1	MMF	✓	✓	✓	✓	✓	✓	✓	7
2	AKI	✓	✓	✓	✓	✓	✓	✓	7
3	Tacrolimus	✓	✓			✓		✓	4
4	1st hospitalization		✓	✓	✓	✓	✓		4
5	3-months eGFR		✓	✓	✓	✓	✓		4
6	DGF	✓			✓			✓	4
7	Donor age			✓	✓	✓	✓		3
8	AR			✓	✓		✓		3
9	Readmissions				✓		✓		3
10	Infection	✓						✓	3
11	Induction	✓						✓	3
12	Hypertension			✓			✓		3
13	Diabetes	✓						✓	3
14	Blood Compatibility	✓						✓	3
15	ERSD Pre-KT			✓		✓	✓		2
16	Transfusion	✓		✓					2
17	Azathioprine				✓			✓	2
18	HLA MM-DR		✓			✓			1
19	Kidney disease		✓			✓			1
20	WIT						✓		1
21	CMV			✓					1
22	Hypertension post KT							✓	1
23	Donor type		✓						1
24	Urological complication	✓							1
25	CIT					✓			1
26	Recipient age				✓				1
27	HLA MM-B								0
28	HLA MM-A								0
29	Gender-Match								0
30	Recipient gender								0
31	Donor gender								0
32	Dialysis modality								0
33	Recipient blood-type								0
34	Donor blood-type								0
35	Cytotoxic antibodies								0

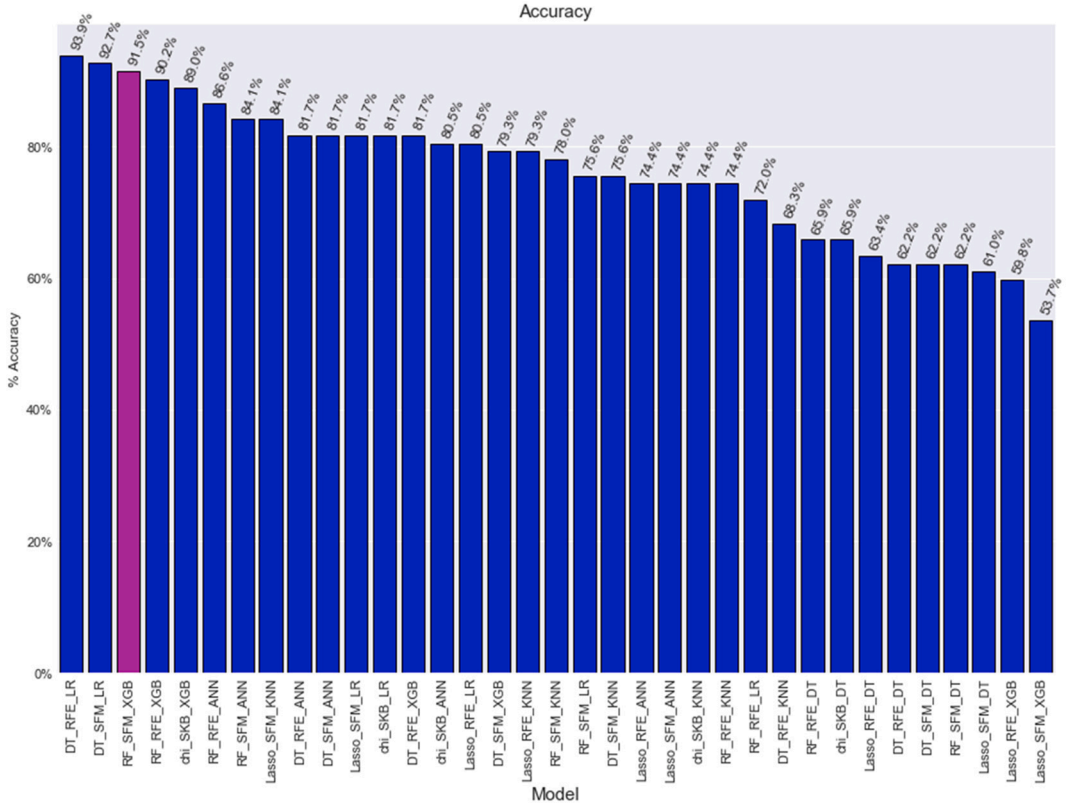


Fig. 4. Accuracy levels of the developed models.

effects on graft survival, which will be discussed later in this section in detail.

The ten features obtained from the RF_SFM method (highlighted in gray in Table 1) result a model with the best performance. About half of these features are among the most frequently selected features by other feature selection methods. They are selected by the other feature selection methods except CMV. They also represent a different category of donor and recipient information such as demographic information, comorbidities, immunosuppressive treatment, and KT procedure. They combine information on pre-transplant, *peri*-transplant, and post-transplant period. This shows the robustness of our analysis and increases the confidence in the selected features.

5.2. Prediction results

This section presents the prediction performance of the five training algorithms. Each one of the seven data groups resulted from the feature selection stage is trained and tested using five algorithms. In other words, we train and test 35 models. The accuracy, sensitivity, specificity, f1-score, and AUC of the testing stage are Figs. 4–8, respectively. The x-axes for all these figures denote the proposed models, while the y-axes are the designated performance measure. Each label in an x-axis can be read as follows: feature selection method, the function used to implement it, and then the algorithm used to build a model. For example, in Fig. 4, the first model from the left is DT_RFE_LR, which means that the features of this model are selected by DT and it is implemented by RFE function, and finally, the model is built using LR. The same goes for all Figs. 4–8.

Six models result in AUCs over 80% and they are built using LR, ANN, and XGB (Fig. 8). More specifically, LR models are based on four data groups obtained from DT_RFE, Chi-sq_SKB, lasso-LR_RFE, and RF_SFM and their AUCs are (85.4%), (84.2%), (83.6%), and (80.9%), respectively. ANN results in an AUC of 80% and it is based on the data group obtained from SFM_RF. XGB model results in the highest AUC (89.7%) and it is based on the data group obtained from RF_SFM. The accuracy, sensitivity, specificity, and F1 of the best model are which are 91.5%, 91.9%, 87.5%, and 89.6% respectively. The performance measures of the best model (RF_SFM_XGB) are highlighted in purple in Figs. 4–8. The F1 score of the best model is the highest comparing with the remaining models.

Other models result in higher sensitivity, specificity, and accuracy than the best model. For example, the LR model that is built based on the data group obtained from DT_SFM provides the highest sensitivity (100%) (See Fig. 5). This model has a better ability than the best model in predicting class one (survived graft). Also, the DT model that is based on the data group obtained from Lasso-LR_SFM results in the highest specificity (100%) (See Fig. 6). This model can correctly predict all cases with zero class (failed graft). However, if we consider the balance between the two classes, the model with the highest AUC is the best, which is the XGB model with the features selected by RF_SFM. In terms of accuracy, the highest accuracy is 93.9%. It is for the LR model that is based on the data group obtained from DT_RFE. However, since our data is imbalanced, the accuracy is a biased measure. In summary, although there are models perform better than the final best model with respect to few measures, but those models perform better in one measure and worse in other measures.

Our final and best model (RF_SFM_XGB) outperforms the models proposed in the literature, which had AUCs ranged from 76% to 88%. The ANN and SVM models proposed by Nematollahi et al. (2017) had AUCs of 86.5% and 76.9%, respectively. A tree-based model proposed by Krikov et al. (2007) resulted in an AUC value of 71.7%. The highest performance model was proposed by Akl et al. (2008). It had an AUC, accuracy, sensitivity, and specificity of 88%, 95%, 88.4%, and 73.2, respectively. However, their model was trained on KT from only living donors. Therefore, their model cannot be used to predict graft survival when KT is performed from a deceased donor.

If we look at our prediction results from the perspective of only algorithms that are used to build the proposed models regardless of

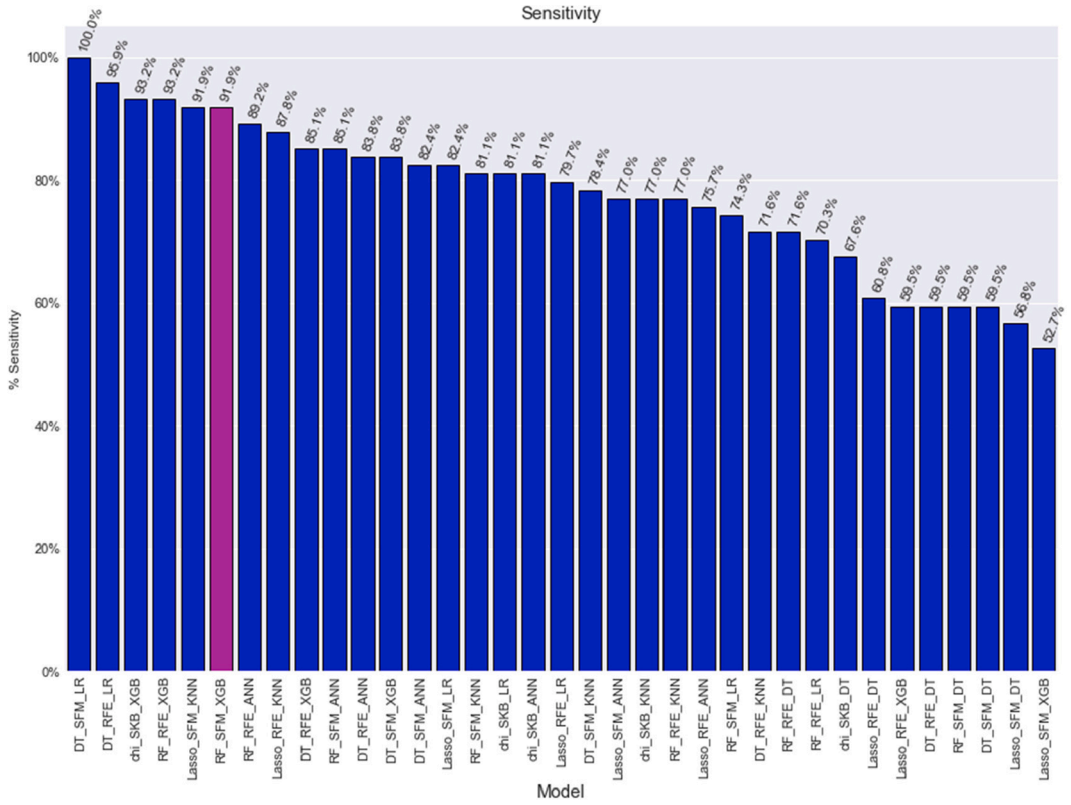


Fig. 5. Sensitivity levels of the developed models.

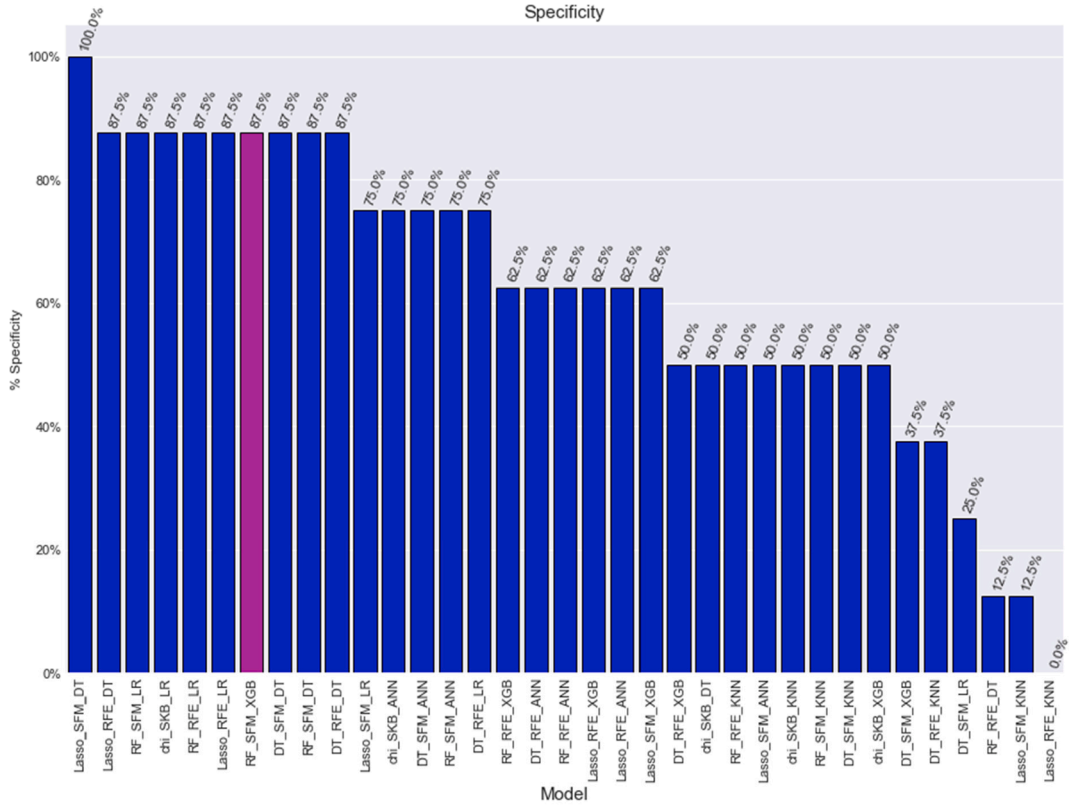


Fig. 6. Specificity levels of the developed models.

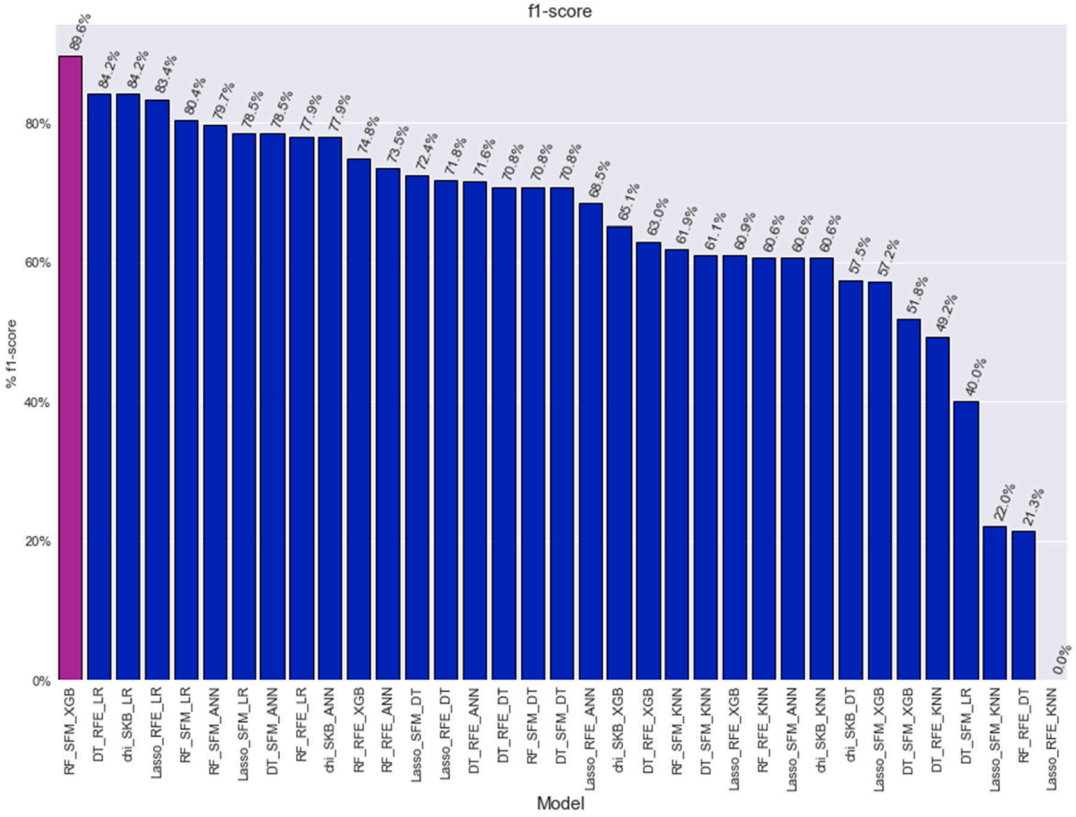


Fig. 7. F1-score for all the proposed models..

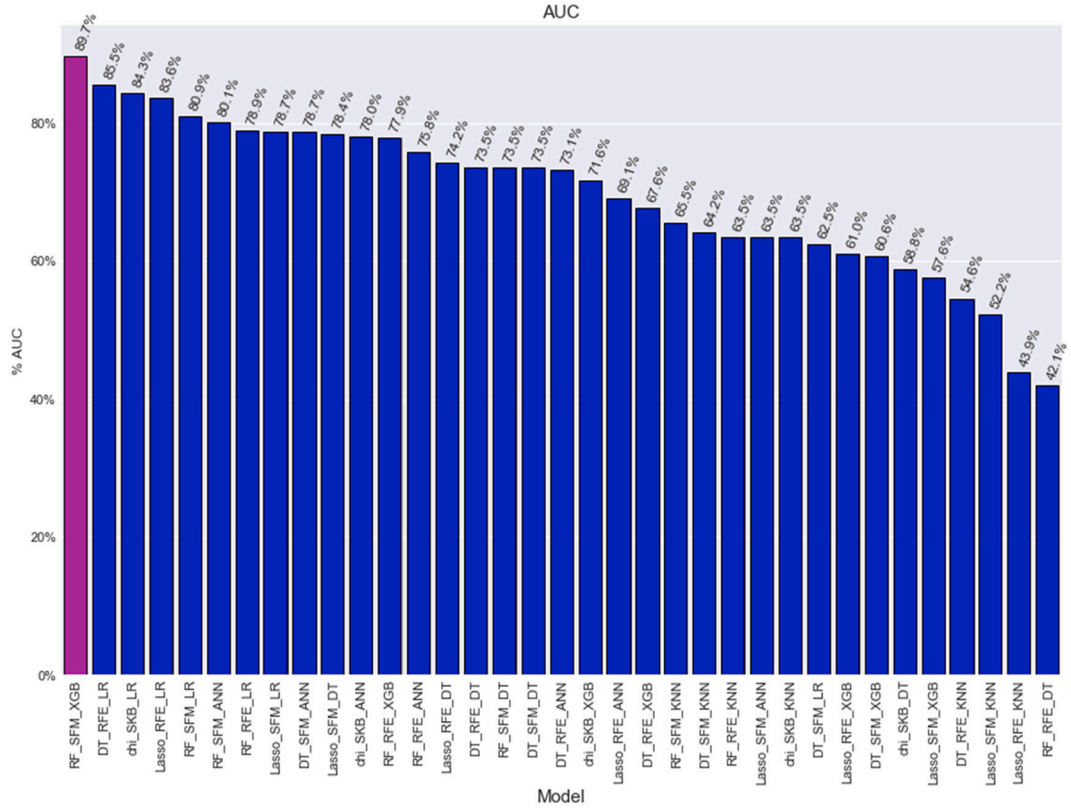


Fig. 8. AUCs of the developed models.

feature selection methods, we can see that XGB results in the best prediction results, followed by LR, ANN, DT, and finally KNN. XGB model outperforms the other four algorithms as it has properties that make it an efficient algorithm. For example, it dynamically decides the depth of decision trees that are used as weak learners by using penalization parameters to prevent high depth trees, which prevents overfitting. It also

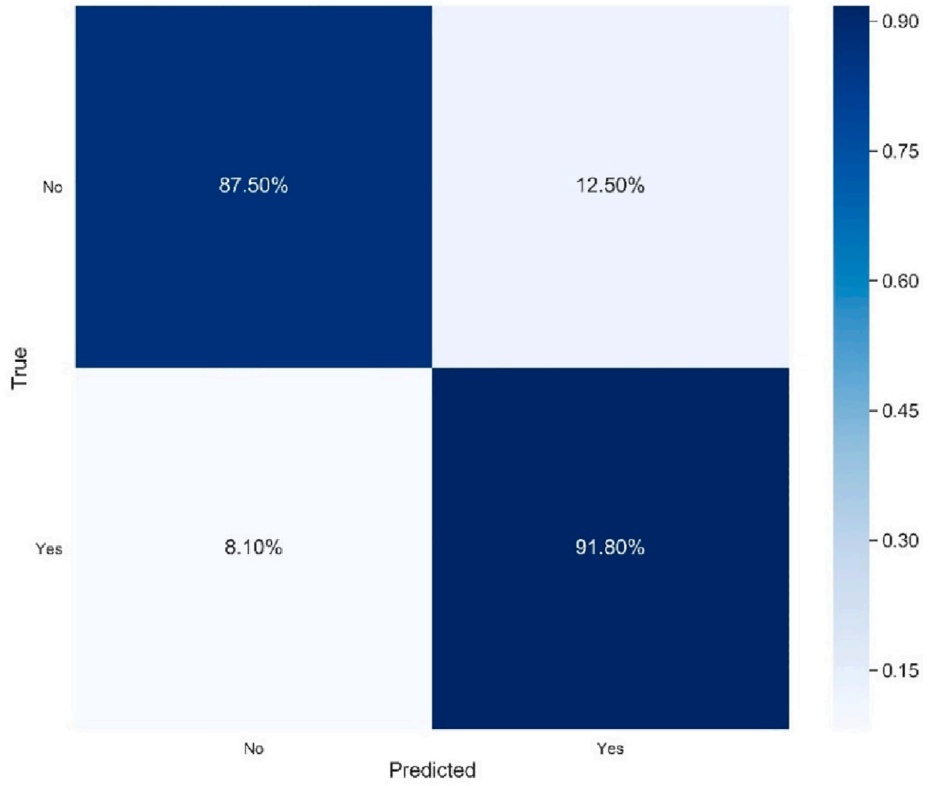


Fig. 9. Confusion matrix of the best model.

adds randomization parameters, which improves optimal learning (Kumar, 2020). For ANN, its performance is worse than XGB and LR, which is because of the size of our database. Since the size is not big, an ANN can be easily either overfitted or under-fitted. Therefore, a huge number of experiments is needed to finetune the ANN parameters. For KNN, it has the worst performance to multiple reasons related to the nature of the algorithm itself. For example, it does not perform well when the number of variables is high. The heterogeneity of our data makes training difficult for the algorithm. Also, the difficulty of determining the optimal number of neighbors is another disadvantage of KNN (Kashyap, 2018).

The confusion matrix of the best model is shown in Fig. 9. The x-axis represents the predicted classes and the y-axis represents the actual classes. The matrix is constructed in a heatmap format, in which the dark color represents the percentage of truly classified instances and the light color represent the percentage of misclassified instances. It can be noticed the model can predict both classes (survival and graft failure). The percentages of truly predicted survived and non-survived graft are 91.80% and 87.50%, respectively. In other words, the rates of TP and TN are high, which makes our model robust and reliable. Fig. 10 shows the feature importance for the best model. It is based on the F-score of the features, which indicates how useful a feature is while building the XGB trees during the model training. The more a feature is used to make a decision in splitting a tree, the higher the F-score.

The performance of the best model can be explained by understanding the effects of the features included in the best model on long-term graft survival. It is not surprising that hypertension is the most important feature. Hypertension is an important risk factor for graft survival (Cheigh, Haschemeyer, Wang, Riggio, Tapia, & Stenzel, 1989; Mange, Cizman, Joffe, & Feldman, 2000; Schwenger, Zeier, & Ritz, 2001). It is a major reason for cardiovascular events (Ojo, 2006), which are known to be the most common cause of graft failure (Ojo et al., 2000). Hence, intensive blood pressure control is recommended in KT recipients (Pagonas et al., 2019).

For blood transfusion, it has been reported by many researchers as one of the most frequent sensitizing factors in patients awaiting KT (Sanfilippo, Vaughn, Bollinger, & Spees, 1982; Scornik et al., 2013). It increases the possibility of graft rejection (Macdougall & Obrador, 2013). AKI is another feature in the best model. It is known as a major factor that affects graft survival in KT patients (Camargo-Salamanca, Garcia-Lopez, Patino-Jaramillo, & Giron-Luque, 2020; Filiponi et al., 2015; Nagarajan et al., 2017). Various risk factors contribute to the occurrence of an AKI episode such as the use of calcineurin inhibitors, chronic kidney disease with a single kidney, ischemia–reperfusion injury, and surgical complications (Nakamura, 2014). Mehrotra et al. (2012) studied the relationship between AKI episodes and graft dysfunction among 3066 KT recipients. It was found that AKI was associated with increased graft loss six months or later after KT. AKI causes incomplete recovery of renal functions from AKI (Nakamura, 2014). It also a risk factor that leads to the progression of chronic kidney disease (Nagarajan et al., 2017). Regarding AR, it is a major risk factor of graft loss (Koo et al., 2015). Patients with a history of AR has poorer long-term renal allograft survival and a higher risk of chronic graft dysfunction than patients with no history of AR (Jalalzadeh, Mousavi-nasab, & Peyrovi, 2015). The same finding is also confirmed by Pallardó Mateu et al. (2004) when they studied the impact of AR episodes on long-term graft survival among 3365 KT recipients.

CMV infection is among the features of the best model. It is a common viral complication after KT (López-Oliva et al., 2017). CMV replication after renal transplant is a risk factor that contributes to long-term graft loss (López-Oliva et al., 2017). CMV infection increases the rate of AR (Borchers, Perez, Kaysen, Ansari, & Gershwin, 1999; Reinke, Prösch, Kern, & Volk, 1999). The relationship between CMV and AR was also pointed out by Nett, Heisey, Fernandez, Sollinger, and Pirsch (2004) and Humar et al. (1999). To mitigate the effects of CMV infection on long-term KT outcomes, some researchers recommended a CMV prophylaxis or preemptive therapy (Singh, 2001). The length of the first hospital stay is another feature in the best model. Hospital recovery after KT

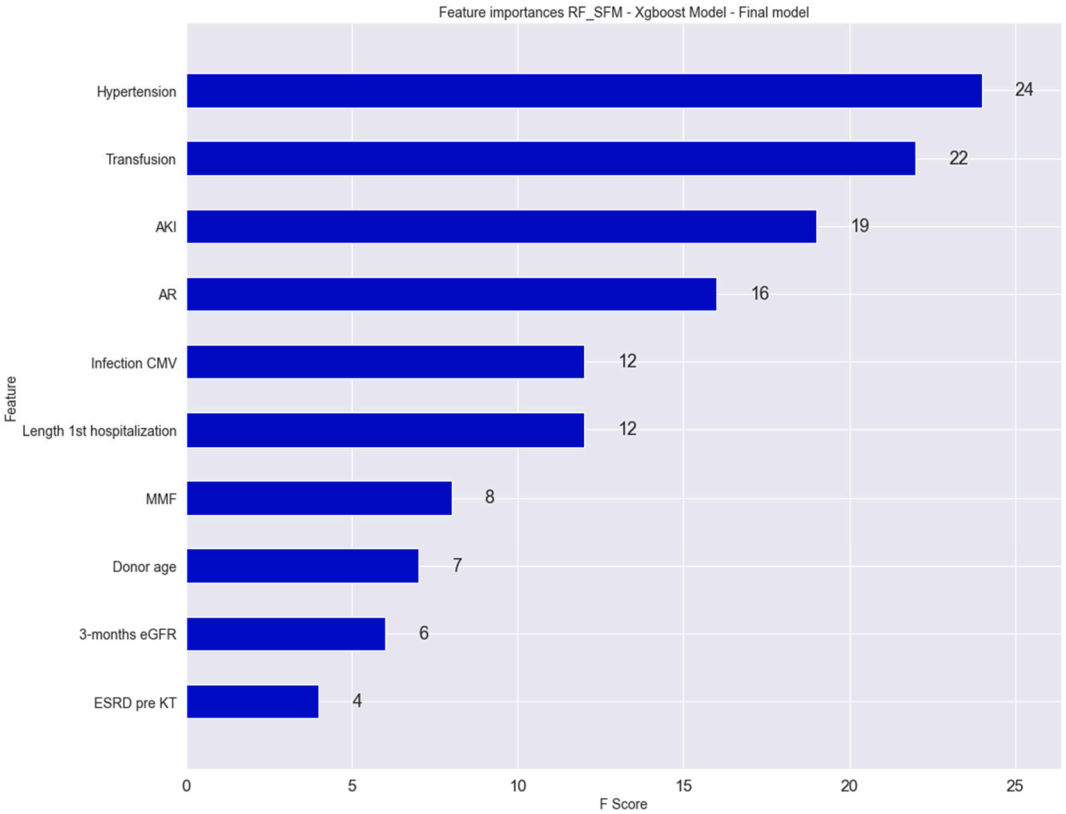


Fig. 10. Feature importance for the final model.

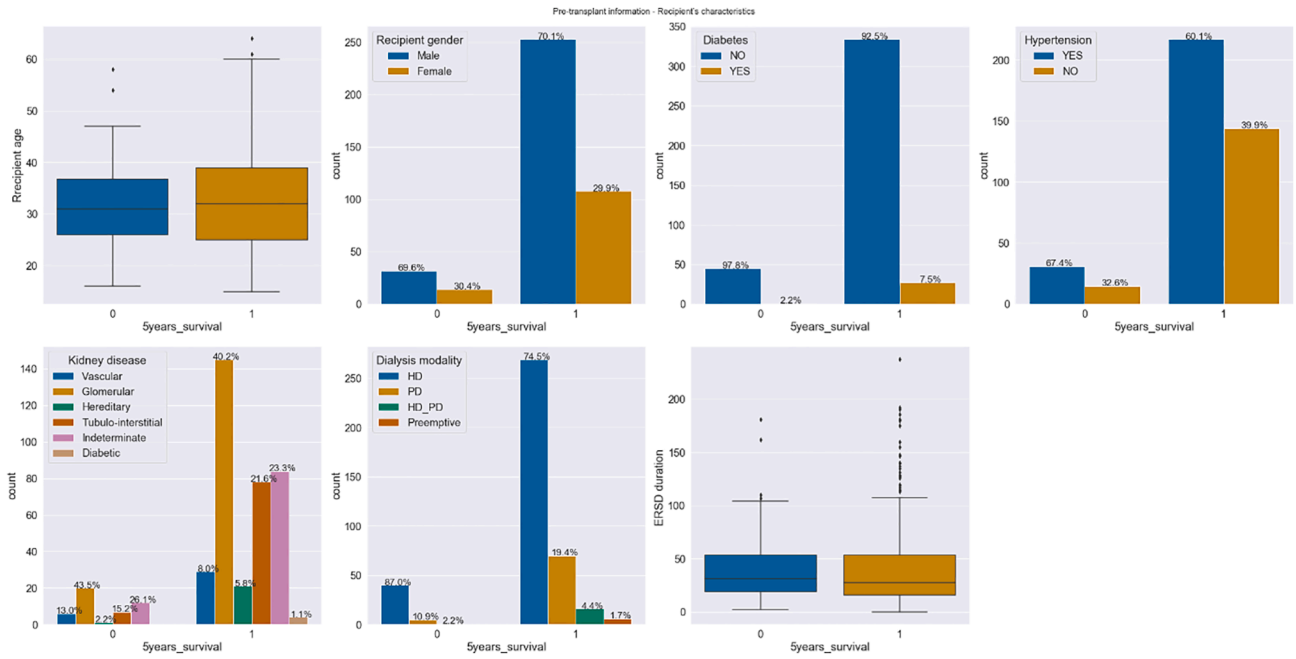


Fig. A1. Pre-transplantation variables – recipient information.

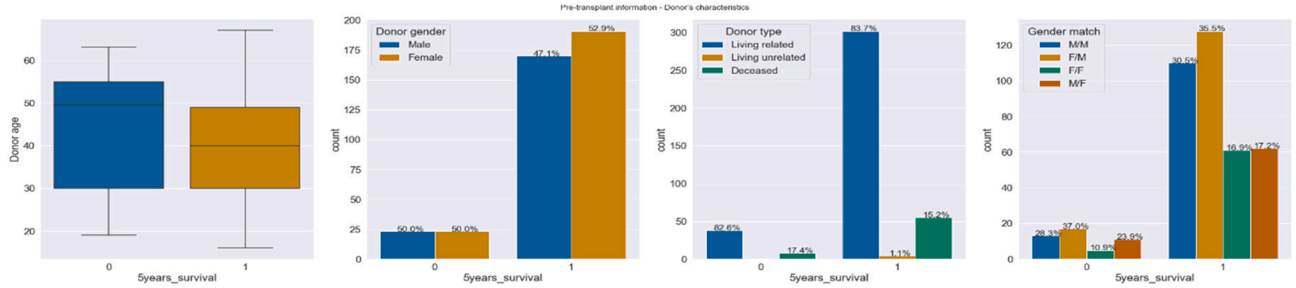


Fig. A2. Pre-transplantation variables – Donor information.

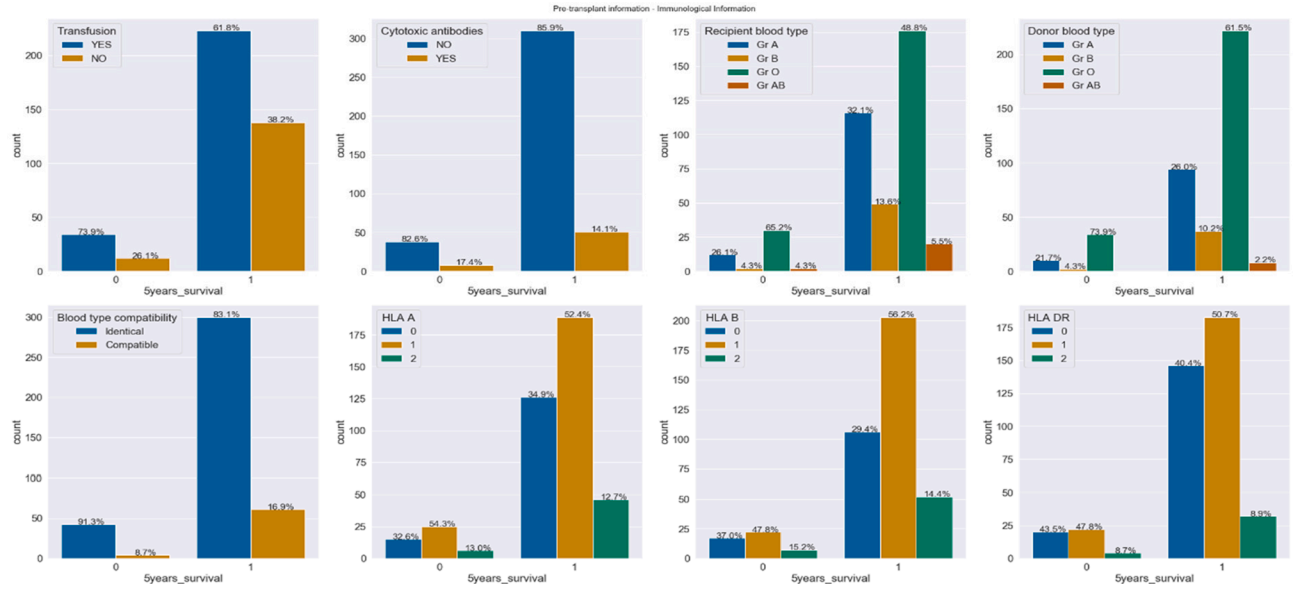


Fig. A3. Pre-transplantation variables – Immunological information.

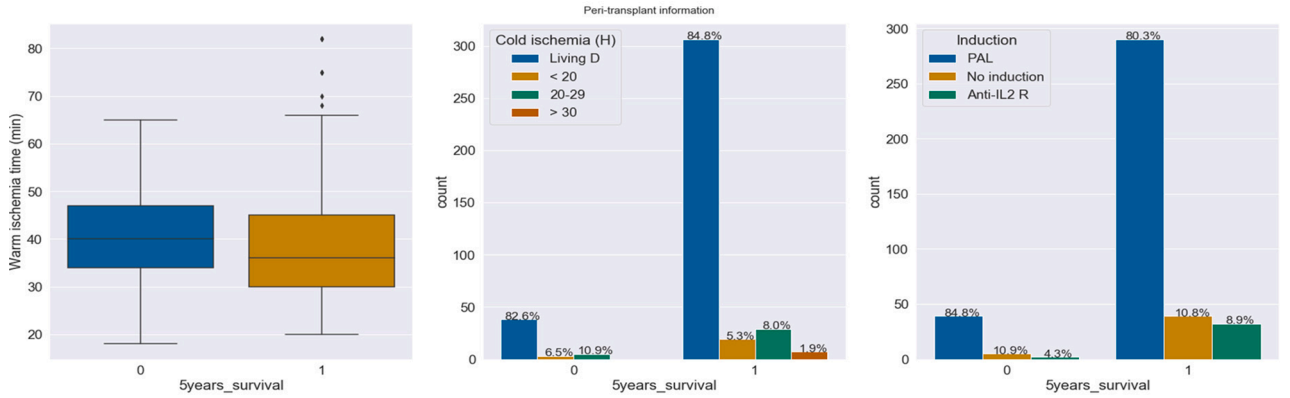


Fig. A4. Peri-transplantation variables.

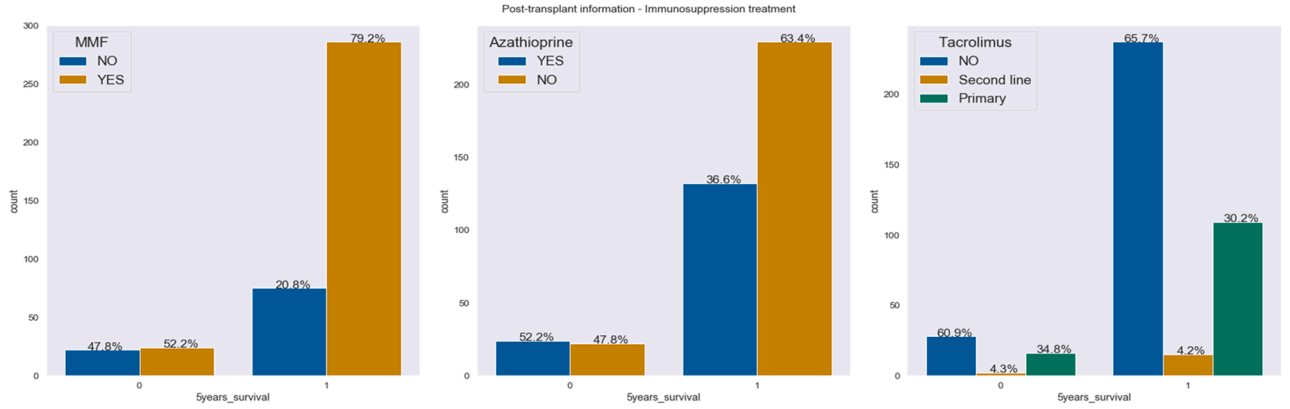


Fig. A5. Post-transplant information – Immunosuppression treatment.

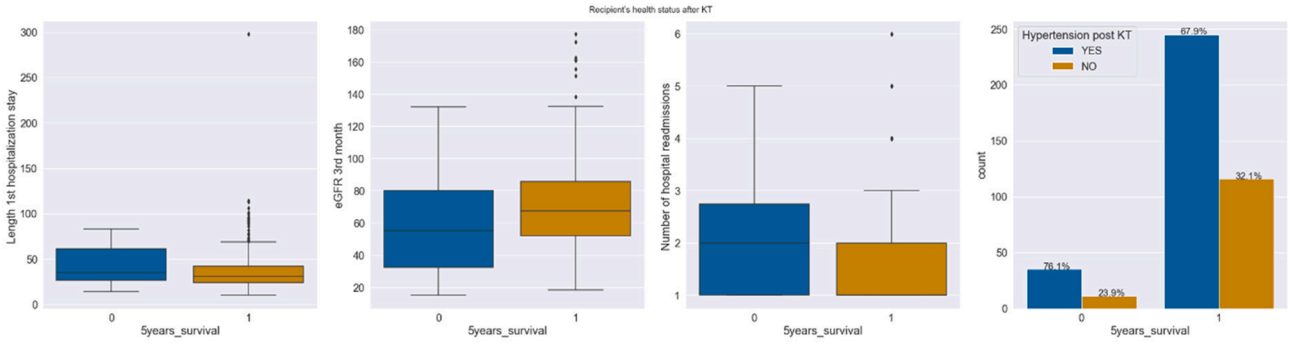


Fig. A6. Recipient's health status after KT.

takes 5–10 days if there are no complications. However, the recovery time increases when complications such as AR, AKI, and CMV infection occur. Donor's age is a feature in the best model as well. It was confirmed in the literature that the donor's age is among the significant factors that affect KT outcomes (Oppenheimer & Aljama, 2004). The older a donor is, the shorter is the length of graft survival. A kidney transplant from elderly donors is associated with many risk factors such as loss of functional renal reserve in old kidneys, hypertension, which is more frequent in the elderly (Lionakis, Mendrinos, Sanidas, Favatas, & Georgopoulou, 2012), and susceptibility to anti-calcineurin caused nephrotoxicity (Moreso et al., 1999). Kidneys from elderly donors are more immunogenic, which increases the risk of AR (de Fijter et al., 2001).

MMF therapy is also selected among the features of the best model. It is used in the majority of renal transplant immunosuppressive regimens. It decreases the chance of AR and consequently improves renal graft survival comparing with other therapies such as azathioprine (Laftavi et al., 2011). The 3-month-eGFR feature is a feature in the best model. It is an indicator that is used to assess the progression of kidney disease in KT recipients (Marcen et al., 2010). When the 3-months $eGFR_{MDRD}$ is less than 30 mL/min/1.73 m^2, it is considered a strong predictor of return to dialysis (Mottola et al., 2020). Finally, the length of the ESRD period before KT is among the features of the best model. In general, ESRD patients would take a course of hemodialysis, peritoneal dialysis, or both, before KT. The longer a patient stays on dialysis before KT, the more likely of resulting negative KT outcome (Bleyer, Burkart, Russell, & Adams, 1999). ESRD and dialysis complications such as phosphocalcic metabolism disorders, chronic inflammation, and anemia promote accelerated atherosclerosis and cardiovascular events (Kon, Linton, & Fazio, 2011). Hence, kidney patients tend to opt-in for transplantation as

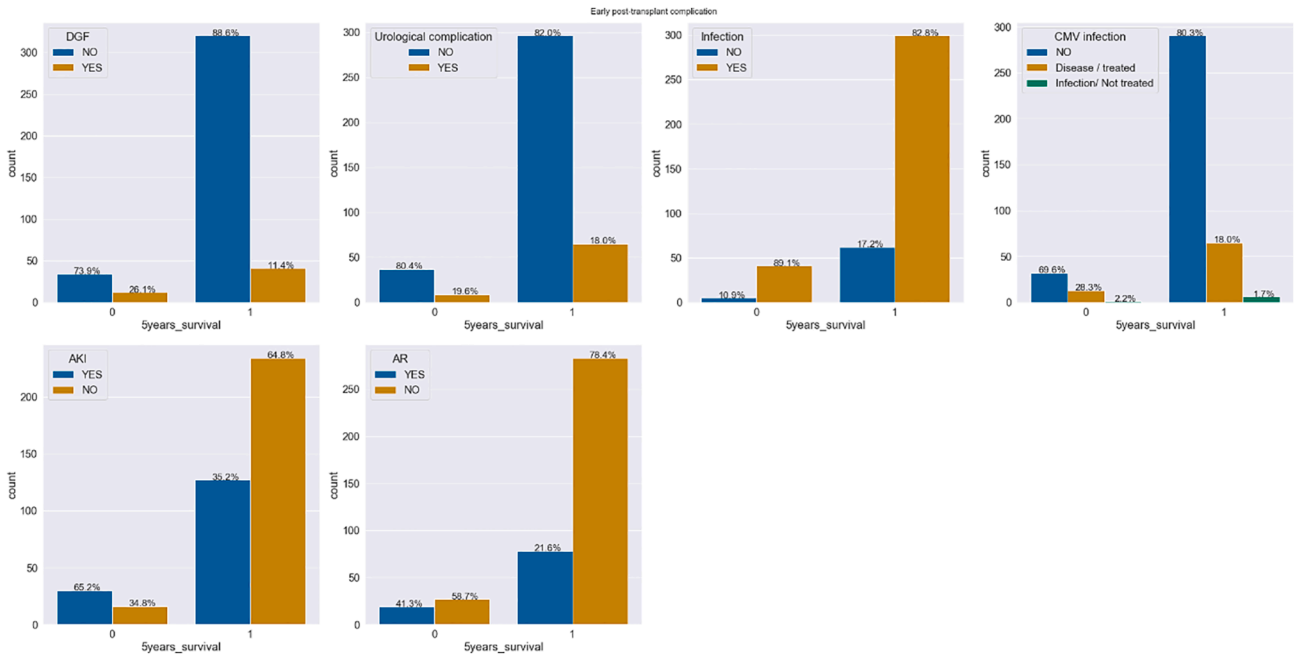


Fig. A7. Early post-transplant complication features.

early as the first ESRD treatment.

It can be noticed that the best model includes ten features, four of them can be obtained before transplant and six of them can be obtained after transplant. The pre-transplant features are donor age, hypertension, ERSD Pre-KT, and transfusion, while post-transplant features are MMF, AKI, 1st hospitalization, 3-month eGFR, AR, and CMV. All those six post-transplant features can be obtained anytime during the first three months except 3-month eGFR feature, which needs at least three months to be obtained. Therefore, our proposed model can used at any time point after three months to predict long-term graft survival. Therefore, it can be used as a decision tool for nephrologists to make daily clinical, therapeutic, and counselling decisions for patients. Suppose that the proposed best model predicts a patient's graft will be lost, a doctor may need to investigate ways to improve the chance of survival. For example, if the patient has a history of CMV infection, the doctor will recommend a reduction in the dose of immunosuppressive treatment and demand a regular monitoring of serum drug concentration.

6. Conclusion and future work

In this study, we proposed a novel approach to predict long-term graft survival after KT. We developed 35 models using seven different feature selection techniques and five predictive modeling algorithms. The feature selection step explained all the factors that affect KT. The features that were selected by most of the seven-feature selection techniques were the most important features. On the other hand, the features that were not selected by any feature selection method were non-important. XGB algorithm resulted in the best model with an AUC, accuracy, sensitivity, specificity, and F1score of 89.7%, 91.5%, 91.9%, 87.5%, and 89.6% respectively. Our best model outperformed the models proposed in the literature with more comprehensive information about the KT process. The best model included ten features: Hypertension, transfusion, AKI, AR, CMV infection, length of the first hospitalization stay, MMF therapy, donor age, 3-months eGFR, and ESRD duration before KT. Those features were identified as the most important features that affect KT as they resulted in the highest AUC. They represent various aspects of pre and post KT. We plan to continue our research and collect additional KT cases from other KT centers in Tunisia to perform external validation. This will confirm the consistency and robustness of our final model before creating a decision support system that can be used by medical practitioners. We also plan to collect more data at the international level to create a more generalized model.

CRediT authorship contribution statement

Samarra Badrouchi: Conceptualization, Investigation, Resources, Data Curation and preprocessing, Writing - original draft (60% of the draft). **Abdulaziz Ahmed:** Methodology, Software, Formal analysis, Writing - original draft (40% of the draft), Writing - review & editing, Visualization, Supervision. **Mohamed Mongi Bacha:** Conceptualization, Resources, Data Curation, Supervision. **Ezzedine Abderrahim:** Data Curation, Resources, Supervision. **Taieb Ben Abdallah:** Investigation, Resources, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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